

Some review questions - 11th week

Q1 Is the projection onto the x -axis, $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, a linear transformation? (Note that for all $(x, y) \in \mathbb{R}^2$, $T((x, y)) = (x, 0)$.) If your answer is no, prove it; if your answer is yes, find the matrix $A \in \mathbb{R}^{2 \times 2}$ such that for all $z \in \mathbb{R}^2$, $T(z) = Az$.

Q2 Is the orthogonal projection $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ onto the line $S = \{(x_1, x_2) \mid x_1 + x_2 = 0\}$ a linear transformation? If your answer is no, prove it; if your answer is yes, find the matrix $A \in \mathbb{R}^{2 \times 2}$ such that for all $z \in \mathbb{R}^2$, $T(z) = Az$.

Q3 Given two matrices $A, B \in \mathbb{R}^{n \times n}$, give a geometric interpretation of the relation $\det(AB) = \det(A)\det(B)$.

Q4 Show that the mapping $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ defined by $f(X) = 2X$ for all $X \in \mathbb{R}^{m \times n}$ is linear.

Q5 Consider the map defined on the set of invertible $n \times n$ matrices by $f(A) = A^{-1}$. Is this mapping a linear transformation? Justify your answer.

Q6 Is the mapping $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{n \times m}$ defined by $f(X) = X^T$ for all $X \in \mathbb{R}^{m \times n}$ a linear transformation? Justify your answer.

Q7 Give the definitions of the sets of functions $\mathcal{C}(\mathbb{R})$ and $\mathcal{C}^1(\mathbb{R})$.

Q8 Consider the mapping $T : \mathcal{C}^1(\mathbb{R}) \rightarrow \mathcal{C}(\mathbb{R})$ defined, for all $f \in \mathcal{C}^1(\mathbb{R})$, by $T(f) = f'$ (the derivative of f also denoted $f'(x) = \frac{df}{dx}$). Is this mapping a linear transformation? If your answer is yes, justify it.

- Q9** State the result on unique representation with respect to a basis.
- Q10** Give the steps of the general algorithm for finding a matrix representation of a linear transformation $f : V \rightarrow W$ when V and W both admit a finite basis.
- Q11** After specifying the bases you use, give a matrix representation of the mapping $T : P_2 \rightarrow P_1$ defined, for all $f \in P_2$, by $T(f) = f'(x)$ (the derivative of f also denoted $f'(x) = \frac{df}{dx}$). Provide the intermediate steps of computation, specifying precisely the bases you chose for the representation.
- Q12** Give a matrix representation of the mapping $T : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ defined by $T(X) = X^T$ for all $X \in \mathbb{R}^{2 \times 2}$. Provide the intermediate steps of computation, specifying precisely the basis you chose for the representation.